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import numpy as np

c1=[1,1,1,1]
c2=[1,-1,1,-1]
c3=[1,1,-1,-1]
c4=[1,-1,-1,1]
rc=[]

print("Enter the data bits :")
d1=int(input("Enter D1 :"))
d2=int(input("Enter D2 :"))
d3=int(input("Enter D3 :"))
d4=int(input("Enter D4 :"))

r1=np.multiply(c1,d1)
r2=np.multiply(c2,d2)
r3=np.multiply(c3,d3)
r4=np.multiply(c4,d4)

resultant_channel=r1+r2+r3+r4;

print("Resultant Channel",resultant_channel)

Channel=int(input("Enter the station to listen for C1=1 ,C2=2, C3=3 C4=4 : "))

if Channel==1:rc=c1
elif Channel==2:rc=c2
elif Channel==3:rc=c3
elif Channel==4:rc=c4

inner_product=np.multiply(resultant_channel,rc)

print("Inner Product",inner_product)

res1=sum(inner_product)

data=res1/len(inner_product)

print("Data bit that was sent",data)

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